

Navin Sridhar

CONTACT INFO	Department of Physics and KIPAC, 452 Lomita Mall, Stanford, CA 94305 E-mail: nsridhar@stanford.edu
EMPLOYMENT	Stanford University, CA • Simons Collaboration (SCEECs) postdoctoral fellowship/KIPAC (2024–2027) • Supervisor: Dr. Roger Blandford
EDUCATION	Columbia University, NY • Ph.D. in Astrophysics (May 2024) • M.A. (2020), M.Phil (2021) in Astrophysics Advisors: Dr. Lorenzo Sironi, Dr. Brian D. Metzger Thesis: Electromagnetic and Multimessenger Signals From Magnetized Outflows of Compact Object Binaries Indian Institute of Science Education & Research, Bhopal • BS-MS in Physics (CPI: 9.05/10) (May 2018) Thesis: Accretion disk evolution around black holes and thermonuclear bursting neutron stars. (Caltech and TIFR)
AWARDS AND FELLOWSHIPS	1. PI of NASA NICER grant (US\$45,000) (2026–2027) 2. Neil Gehrels Prize Fellowship: JSI, University of Maryland & NASA (2024) 3. Brinson Fellowship: Northwestern University/CIERA & University of Chicago (2024) 4. NASA <i>FINESST</i> award (US\$100,000) (2022–2024) 5. PI of NASA NICER grant (US\$37,500) (2022–2024) 6. Simons Foundation Aspen center for physics grant (US\$1,350) (July 2023) 7. International Astronomical Union (S378) travel grant (US\$1,600) (March 2023) 8. Columbia Arts and Sciences Graduate Council travel grant (US\$500) (2022) 9. Columbia ASGC Student/Diversity Initiative Grant (US\$2,000) (2022–2023) 10. AAS National Osterbrock Leadership fellow (US\$6,400) (2021–2023) 11. The Yeh Family endowment fellowship, Columbia University (2019–2020) 12. Dean’s fellowship for graduate study at Columbia University (2018–2020) 13. Caltech Summer Undergraduate Research Fellowship (US\$6,600) (July–Oct 2017) 14. Mitacs Globalink Research Internship Award (~CA\$6,600) (May–July 2017) 15. <i>INSPIRE</i> fellowship by DST, Govt. of India (INR 400,000) (2013–2018)
RESEARCH EXPERIENCE/ VISITS	• Affiliated scientist, SCEECs (2023–2028) • Simons Collaboration on Extreme Electrodynamics of Compact Sources • Visiting Researcher, KITP, UC Santa Barbara (Sept 2024) • Visiting Student Researcher, Caltech, Pasadena (2023–2024) Host: Dr. Sterl Phinney • Stellar Interactions and the Transients they Cause, Aspen (July 2023) Host: Aspen Center for Physics, Colorado • Visiting researcher, Simons Foundation – plasma astrophysics program (2019–2023) Center for Computational Astrophysics, Flatiron Institute, NY Advisor(s): Dr. Sasha Philippov, Dr. Lorenzo Sironi • Summer Undergraduate Research Fellow (SURF), Caltech, Pasadena (July–Oct 2017) Advisor(s): Dr. Javier García, Prof. Fiona Harrison • Mitacs Globalink Research Intern, University of Alberta, Edmonton (May–July 2017) Advisor: Dr. Rodrigo Fernández • Visiting Summer Research Program, TIFR, Mumbai (May–July 2016) Advisor: Prof. Sudip Bhattacharyya • Indian Academy of Sciences summer research fellow, ISRO, Bangalore (June–Aug 2015) Advisor: Dr. Anuj Nandi • Visiting Student Program, IITp, Bangalore (May–Dec 2015) Advisor: Prof. Aruna Goswami

1. **Sridhar, N.**, 2026. “*Interaction of Strong X-mode Waves with Magnetized Pair Plasmas*” arXiv:2610.14204, submitted to PRL.
2. **Sridhar, N.**, Sobacchi, E., Sironi, L., et al. 2026. “*Interaction of Strong Electromagnetic Waves with Unmagnetized Pair Plasmas*” Physical Review Letters 137, 045102.
3. **Sridhar, N.**, Ripperda, B., Sironi, L., et al. 2025. “*Bulk Motions in the Black Hole Jet Sheath as a Candidate for the Comptonizing Corona*” ApJ 979, 199
4. **Sridhar, N.**, Metzger, B. D., & Fang, K. 2024. “*High-Energy Neutrinos from Gamma-Ray-Faint Accretion-Powered Hypernebulæ.*” ApJ 960, 74
5. **Sridhar, N.**, Sironi, L., & Beloborodov, A. M. 2023. “*Comptonization by Reconnection Plasmoids in Black Hole Coronæ II: Electron-Ion Plasma.*” MNRAS 518, 1301-1315
6. **Sridhar, N.**, & Metzger, B. D. 2022. “*Radio Nebulæ from Hyper-Accreting X-ray Binaries as Common Envelope Precursors and Persistent Counterparts of FRBs*” ApJ 937, 5
7. **Sridhar, N.**, Sironi, L., & Beloborodov, A. M. 2021. “*Comptonization by Reconnection Plasmoids in Black Hole Coronæ I: Magnetically Dominated Pair Plasma*” MNRAS 507, 5625-5640
8. **Sridhar, N.**, Metzger, B. D., Beniamini, Paz., et al. 2021. “*Periodic Fast Radio Bursts from Luminous X-ray Binaries.*” ApJ 917, 13
9. **Sridhar, N.**, Zrake, J., Metzger, B. D., et al. 2021. “*Shock-powered radio precursors of neutron star mergers from accelerating relativistic binary winds.*” MNRAS 501, 3184-3202
10. **Sridhar, N.**, García, J. A., Steiner, J. F., et al. 2020. “*Evolution of the Accretion Disk—Corona during the Bright Hard-to-soft State Transition: A Reflection Spectroscopic Study with GX 339-4.*” ApJ 890, 53
11. **Sridhar, N.**, Bhattacharyya, S., Chandra, S., & Antia, H. M. 2019. “*Broad-band reflection spectroscopy of MAXI J1535-571 using AstroSat: estimation of black hole mass and spin.*” MNRAS 487, 4221-4229

≤ 3rd author, with significant contributions (11): [[†] = supervised student]

12. Fan, N.[†], & **Sridhar, N.** 2026. “*Comptonization by Shear-driven Bulk Motions in Black Hole Coronæ*” to be submitted to ApJ
13. Uno, K.[†], Metzger, B. D., **Sridhar, N.** et al. 2026. “*The Population of Radio Hypernebulæ from Mass-Transferring Massive Binaries*” to be submitted to ApJ
14. Gupta, S.[†], **Sridhar, N.**, Sironi, L. 2025. “*The Role of Electric Dominance for Particle Injection in Relativistic Reconnection*” MNRAS 538, 49-59
15. Gupta, S.[†], **Sridhar, N.**, Sironi, L. 2023. “*Comptonization by Reconnection Plasmoids in Black Hole Coronæ III: Dependence on the Guide Field in Pair Plasma.*” MNRAS 527, 6065-6075
16. Bhandari, S., Marcote, B., **Sridhar, N.**, et al. 2023. “*Constraints on the persistent radio source associated with FRB 20190520B using the European VLBI Network*”, ApJ 958, 2
17. Eftekhari, T., Fong, W-F., Gordon, A. C., **Sridhar, N.**, et al. 2023. “*An X-ray census of Fast Radio Burst host galaxies: constraints on AGN and X-ray counterparts*”, ApJ 958, 1
18. Metzger, B. D., **Sridhar, N.**, Margalit, B., et al. 2022. “*A Toy Model for the Time-Frequency Structure of Fast Radio Bursts: Implications for the CHIME/FRB Burst Dichotomy.*” ApJ 925, 135
19. Margalit, B., Beniamini, P., **Sridhar, N.**, Metzger, B. D., 2020. “*Implications of a Fast Radio Burst from a Galactic Magnetar.*” ApJL 899, L27
20. García, J. A., Tomsick, J. A., **Sridhar, N.**, et al. 2019. “*The 2017 Failed Outburst of GX 339-4: Relativistic X-Ray Reflection near the Black Hole Revealed by NuSTAR and Swift Spectroscopy.*” ApJ 885, 48
21. Bhattacharyya, S., Yadav, J. S., **Sridhar, N.**, et al. 2018. “*Effects of Thermonuclear X-Ray Bursts on Non-burst Emissions in the Soft State of 4U 1728-34.*” ApJ 860, 88
22. Karinkuzhi, D., Goswami, A., **Sridhar, N.**, et al. 2018. “*Chemical analysis of three barium stars: HD 51959, HD 88035, and HD 121447.*” MNRAS 476, 3086-3096

> 3rd author (19):

23. Bhattacharyya, S., Dinsmore, J., ..., **Sridhar, N.**, et al. 2026. “*High-quality optical pulse profile and simultaneous X-ray pulsations from an accreting millisecond X-ray pulsar*” arXiv:2700.00000; Submitted to ApJ.

24. Adegoke, O., Bianchi, M., ..., **Sridhar, N.**, et al. 2026. “*Following the Flow: Accretion and X-Ray Reflection in the 2021 Outburst of GX 339-4*” arXiv:2700.00000; Submitted to ApJ.
 25. Shuvo, P., Meyer, E., ..., **Sridhar, N.**, et al. 2026. “*First Detection of Radio Polarization During Jet Formation in the Changing-Look AGN 1ES 1927+654*” arXiv:2609.09539; Accepted in the ApJ.
 26. Uttarkar, P., Shannon, R., ..., **Sridhar, N.**, et al. 2026. “*A fast radio burst cyclone in technicolour: evidence of plasma lensing*” arXiv:2602.16409; Submitted to Nature Astronomy
 27. Moroianu, A., Bhandari, S., ..., **Sridhar, N.**, et al. 2026. “*A Milliarcsecond Localization Associates FRB 20190417A with a Compact Persistent Radio Source and an Extreme Magnetoionic Environment*” ApJL 996, L16
 28. Snelders, M. P., Hessels, J. W. T., ..., **Sridhar, N.**, et al. 2025. “*Revisiting FRB 20121102A: milliarcsecond localisation and a decreasing dispersion measure*” arXiv:2510.11352
 29. Blanchard, P., Berger, E., ..., **Sridhar, N.**, et al. 2025. “*James Webb Space Telescope Observations of the Nearby and Precisely Localized FRB20250316A: A Potential Near-IR Counterpart and Implications for the Progenitors of Fast Radio Bursts*” ApJL 989, L49
 30. Ibik, A., Drout, M., ..., **Sridhar, N.**, et al. 2024. “*A search for persistent radio sources toward repeating fast radio bursts discovered by CHIME/FRB*” ApJ 976, 199
 31. Dong, Y., Eftekhari, T., ..., **Sridhar, N.**, et al. 2024. “*A Radio Study of Persistent Radio Sources in Nearby Dwarf Galaxies: Implications for Fast Radio Bursts*”, ApJ 973, 133
 32. Sarin, N., Clarke, T. A., ..., **Sridhar, N.** 2024. “*The origin of the coherent radio flash potentially associated with GRB 201006A*”, ApJL 973, L20
 33. AXIS TDAMM SWG., ..., incl. **Sridhar, N.**, et al. 2023. “*Prospects for Time-Domain and Multi-Messenger Science with AXIS*”, Universe, 10, 316
 34. Connors, R. M. T., Tomsick, J. T., ..., **Sridhar, N.**, et al. 2023. “*The High Energy X-ray Probe (HEX-P): Probing Accretion onto Stellar Mass Black Holes*”, Front. Astron. Space Sci. 10:1292682
 35. Kammoun, E., Lohfink, A. M., ..., **Sridhar, N.**, et al. 2023. “*The High Energy X-ray Probe (HEX-P): Probing the physics of the X-ray corona in active galactic nuclei*”, Front. Astron. Space Sci. 10:1308056
 36. Dong, Y., Eftekhari, T., ..., **Sridhar, N.**, et al. 2023. “*Mapping Obscured Star Formation in the Host Galaxy of FRB 20201124A*”, ApJ 961, 44
 37. Bhandari, S., Gordon, A. C., ..., **Sridhar, N.**, et al. 2023. “*A non-repeating fast radio burst in a dwarf host galaxy*”, ApJ 948, 67
 38. Connors, R. M. T., García, J. A., ..., **Sridhar, N.**, et al. 2021. “*Reflection Modeling of Black Hole Binary 4U 1630–47: Disk Density and Returning Radiation*”, ApJ 909, 146
 39. Connors, R. M. T., García, J. A., ..., **Sridhar, N.**, et al. 2020. “*Evidence for Returning Disk Radiation in the Black Hole X-Ray Binary XTE J1550–564.*” ApJ 892, 47
 40. Connors, R. M. T., García, J. A., ..., **Sridhar, N.**, et al. 2019. “*Conflicting Disk Inclination Estimates for the Black Hole X-Ray Binary XTE J1550–564.*” ApJ 882, 179
 41. Verdhnan Chauhan, J., Yadav, J. S., ..., **Sridhar, N.**, et al. 2017. “*AstroSat/LAXPC Detection of Millisecond Phenomena in 4U 1728-34.*” ApJ 841, 41
- Instrumentation science white papers (7):
42. De Marco, B., ., incl. **Sridhar, N.**, et al. 2026. “*Mapping the evolving inner regions of the accretion flow around stellar mass black holes with NewAthena* ”
 43. De Grandis, D., ., incl. **Sridhar, N.**, et al. 2026. “*Exploring isolated neutron stars from pulsars to magnetars with NewAthena*”
 44. Safi-Harb, S., ., incl. **Sridhar, N.**, et al. 2026. “*Illuminating Pulsar Wind Nebulae Across Evolutionary Stages with NewAthena*”
 45. Anitra, A., ., incl. **Sridhar, N.**, et al. 2026. “*High-Frequency Phenomena in Neutron Star Low-Mass X-ray Binaries with NewAthena*”
 46. Jaisawal, G., ., incl. **Sridhar, N.**, et al. 2026. “*From Faint Outbursts to Thermonuclear Explosions: unveiling the secrets of Neutron Star LMXBs with NewAthena*”
 47. Koss, M., ., incl. **Sridhar, N.**, et al. 2025. “*The Advanced X-ray Imaging Satellite Community Science Book*”, arXiv:2511.00253

48. Safi-Harb, S., Burdge, K., ..., incl. **Sridhar, N.**, et al. 2023. “*From Stellar Death to Cosmic Revelations: Zooming in on Compact Objects, Relativistic Outflows and Supernova Remnants with AXIS*”, arXiv:2311.07673

Astronomical Telegrams (2):

49. Blanchard, P., Berger, E., ..., **Sridhar, N.**, et al. 2025. “*Deep JWST/NIRCam Imaging of FRB 20250316A: Detection of Potential IR Counterparts.*” ATel #17195
50. García, J. A., Harrison, F., ..., **Sridhar, N.**, et al. 2017. “*NuSTAR Observation of GX 339–4 in the early stages of its 2017 outburst.*” ATel #10825

COMPUTATING
AWARDS

- NASA NICER on Athena AO8 2026-2027: ~ 35 million CPU hours (1.4 million SBUs)
- NASA FINESST on Pleiades 2022-2024: ~ 35 million CPU hours (1 million SBUs)
- NASA NICER on Pleiades AO4 2021-2023: ~ 30 million CPU hours (1.2 million SBUs)

OBSERVING
AWARDS/
EXPERIENCE

X-ray: 15 Ms; Radio: 142 ks; Optical: 65 ks; IR: 4.5 hr

Total telescope grant: \$624,935

As Principal Investigator:

1. NICER archival data of transitional ms pulsar (Cycle AO8; 1.2 Ms of data; US\$45,000).
2. Joint NICER and NuSTAR observations studying reflection features from X-ray binaries (Cycle AO3; 60 ks; US\$37,500).
3. uGMRT multi-band radio observations of Giant Radio Pulses (Cycle 34; 36 ks).
4. *AstroSat* observations of ~10 sources: accretion powered millisecond pulsars, neutron star type-I bursts, black hole X-ray binaries (Cycles A04-A08; cumulatively >300 ks).

As Co-Investigator:

1. JWST search for a transient NIR counterpart to FRB 20250316A (PI: Peter Blanchard; \$110,500 ; cycle 5, 2026).
2. JWST observation of FRB 20250316A (PI: Peter Blanchard; cycle 4, 2025).
3. Joint NuSTAR and NICER observation to study the reflection properties and evolution of accretion disks in black hole binaries (Cycle 10; 500 ks; PI: J. Garcia).
4. GBT observations to probe the dynamic environment of FRB 20180301A (Cycle 2024B; 108 ks; PI: P. Kumar).
5. Chandra observations to probe the engines of fast radio bursts (Cycle 25; 106 ks; PI: T. Eftekhari).
6. GBT observations to probe the local environment of FRB 20180301A (Cycle 2023B; 72 ks; PI: P. Kumar).
7. VLA observations to constrain FRB ‘persistent radio sources’ (Cycle 2023B; 10 ks; PI: Y. Dong).
8. NICER observations for amassing a comprehensive spectral timing archive of outbursting black holes (Cycle AO5; 200 ks; \$43,000; PI: J. Steiner).
9. Joint NICER and NuSTAR observation to study the reflection properties and evolution of accretion disks in black hole binaries (Cycle AO4; 110 ks; \$44,000 ; PI: J. Garcia).
10. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO4; 88 ks; \$44,000; PI: R. Connors).
11. NICER observations for black hole spin measurements using continuum fitting (Cycle AO4; 40 ks; \$37,500; PI: J. Steiner).
12. Joint NICER and NuSTAR observations for measuring black hole spin and mass through X-ray reflection spectra and reverberation lags (Cycle AO4; 30 ks; PI: G. Mastroserio).
13. Joint XMM-Newton and NuSTAR observations for studying truncated accretion disks in black holes (AO 21; 120 ks; \$50,656; PI: J. García).
14. Chandra observation of nearby Fast Radio Bursts (Cycle 23; 55 ks; \$52,519; PI: T. Eftekhari).
15. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO3; 88 ks; \$37,500; PI: R. Connors).
16. NICER observations for black hole spin measurements using continuum fitting (Cycle AO3; 40 ks; \$44,000; PI: J. Steiner).
17. Joint NICER and NuSTAR observations for measuring black hole spin and mass through X-ray reflection spectra and reverberation lags (Cycle AO3; 30 ks; \$37,500; PI: G. Mastroserio).

18. NICER observations monitoring the evolving thermal disk emission of LMC X-3 (Cycle AO2; 60 ks; \$20,229; PI: J. Steiner).
19. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO2; 90 ks; \$21,031; PI: R. Connors).

Others:

1. Mt. Palomar 200" Hale telescope: DBSP observations of 14 variable AGN (2 nights)
2. NRAO Green Bank Telescope: radio pulsar observations (24 ks)

TALKS/

PRESENTATIONS

(*32 INVITED)

86. *Talk at the PCTS workshop on Black Hole corona 2027 **Princeton University**, NJ (March 2027)
85. *Talk at the SCEECS black hole corona meeting 2026 **Princeton University**, NJ (Oct 2026)
84. Talk at the Kavli Astrophysics Symposium 2026 **Cambridge**, UK (Sept 2026)
83. Talk and *discussion session lead at FRB 2026 conference, **Guiyang**, China (July 2026)
82. *Seminar at Department of Astronomy, **Tsinghua University**, Beijing (July 2026)
81. Talk at the BlackHolic 2026 conference, **Oxford University**, UK (March 2026)
80. *Astrophysics seminar, **Washington University**, St. Louis, MO (Oct 2025)
79. 22nd AAS-High Energy Astrophysics Division meeting, St. Louis, MO (Oct 2025)
78. *Black-board talk, Frontiers of Relativistic Plasma Physics, **KITP**, USA (Sep 2025)
77. *Overview of FRB emission mechanisms, FRB 2025 conference, **McGill**, Canada (July 2025)
76. Seminar, **Inter-University Center for Astronomy & Astrophysics**, Pune (June 2025)
75. *Astrophysics colloquium, **Tata Institute of Fundamental Research**, Mumbai (May 2025)
74. Astrophysics and Relativity seminar, **International Center for Theoretical Sciences**, Bangalore (May 2025)
73. Relativistic Magnetospheres workshop, **Ecole De Physique Des Houches**, France (April 2025)
72. Magnetic fields around Compact Objects workshop, **Perimeter Institute**, Canada (March 2025)
71. Fast Radio Burst Frontiers workshop, **Carnegie Mellon University**, USA (March 2025)
70. *Talk at the XOC group meeting **Stanford**, CA (Sept 2024)
69. Talk at the High-energy plasma phenomena in astrophysics workshop **MIAPbP**, Munich, Germany (Sept 2024)
68. Graduating student colloquium, **Columbia University**, NY (April 2024)
67. ***AXIS** seminar series (April 2024)
66. 21st AAS-High Energy Astrophysics Division meeting, Texas (April 2024)
65. *Seminar, Simons Collaboration on Extreme Electrodynamics of Compact Object Sources (**SCEECS**) (Feb 2024)
64. *Astroplasmas seminar, **Princeton University**, NJ (Dec 2023)
63. *CIERA Theory seminar, **Northwestern University**, IL (Nov 2023)
62. JSI workshop on 'Winds throughout the Universe', **Annapolis**, MD (Oct 2023)
61. *Theoretical astrophysics group board talk, **Carnegie Observatories**, CA (Sept 2023)
60. *Seminar, **Syracuse University**, NY (Sept 2023)
59. *Review talk at Astrophysics of Fast Radio Bursts II workshop (+session chair), **CCA, Flatiron Institute**, NY (Sept 2023)
58. *Colloquium, **National Center for Radio Astrophysics**, Pune, India (June 2023)
57. *Astrophysics seminar, **Tata Institute of Fundamental Research**, Mumbai (May 2023)
56. *'Transient Tuesday' talk, DARK, Niels Bohr Institute **Copenhagen**, Denmark (May 2023)
55. *Discussion chair, Fast Radio Burst Follow-Up workshop, **University of Toronto** (April 2023)
54. *High energy astrophysics seminar, **Hebrew University of Jerusalem**, Israel (March 2023)
53. *Seminar, Astrophysics Research Center of **The Open University**, Israel (March 2023)
52. *High energy astrophysics meeting, Black Hole Initiative, **Harvard**, MA (Feb 2023)
51. Cosmic transients lab seminar, Center for Astrophysics, **Harvard**, MA (Feb 2023)
50. High Energy Astrophysics seminar, Center for Astrophysics, **Harvard**, MA (Feb 2023)
49. FRB Taiwan 2023, National Chung Hsing University, Taichung **Taiwan** (Feb 2023)
48. *Workshop on 'Overcoming disconnects in the understanding of accreting black holes', **Lorentz Center, Leiden**, The Netherlands (Jan 2023)
47. HEX-P black hole binaries, **science working group** presentation (Dec 2022)
46. *CCAPP seminar, **Ohio State University**, OH (Nov 2022)
45. 64th APS Division of Plasma Physics annual meeting, **Spokane**, WA (Oct 2022)
44. Review talk on Observations of FRBs, **CCA, Flatiron Institute**, NY (Oct 2022)
43. ***AXIS** 'compact objects and supernova remnants' **science working group** (Oct 2022)
42. *Astrophysics seminar, **Cornell University**, NY (Sept 2022)

41. *AGN seminar at **NASA** Goddard Space Flight Center, MD (Sept 2022)
40. **NASA** Time Domain and Multi-Messenger (TDAMM) Initiative Workshop, **Annapolis**, MD (Aug 2022)
39. FRB 2022 meeting, IAUGA, **Busan**, South Korea (Aug 2022)
38. Summer BLAST workshop, Black Hole Initiative, **Harvard**, MA (July 2021)
37. *AXIS* ‘Time-Domain and Multi-Messenger Astronomy’ **science working group** (July 2022)
36. Special pizza lunch seminar, **Caltech**, CA (June 2022)
35. Seminar at **Carnegie Observatories**, CA (June 2022)
34. 240th meeting of the American Astronomical Society, **Pasadena**, CA (June 2022)
33. Workshop on Relativistic Plasma Astrophysics, **Purdue University**, IN (May 2022)
32. Compact objects group meeting at **CCA, Flatiron Institute**, NY (April 2022)
31. *Kapteyn Astronomical Institute, **Groningen**, The Netherlands (April 2022)
30. 19th AAS-High Energy Astrophysics Division meeting, **Pittsburgh**, PA (March 2022)
29. Brown bag lunch seminar, **Massachusetts Institute of Technology**, MA (Feb 2022)
28. Winter BLAST workshop, Black Hole Initiative, **Harvard**, MA (Dec 2021)
27. Gothamfest, **CCA, Flatiron Institute**, NY (Dec 2021)
26. Kavli-IPMU workshop on particle acceleration by reconnection, **Tokyo**, Japan (Nov 2021)
25. 9th Microquasar workshop, **Cagliari**, Italy (Sept 2021)
24. Astrofest, **Columbia University**, NY (Sept 2021)
23. High-energy plasma phenomena in astrophysics, **Munich**, Germany (July 2021)
22. FRB 2021 conference (online) (July-Aug 2021)
21. *Princeton astro-coffee, **Princeton**, NJ (July 2021)
20. ‘Midwest Magnetic Fields’ Workshop, **Wisconsin-Madison**, (June 2021)
19. *CHIME/FRB collaboration, **McGill**, Canada (May 2021)
18. *Seminar at the **Institute of Astrophysics - FORTH**, Greece (April 2021)
17. Compact objects group meeting at **CCA, Flatiron Institute**, NY (March 2021)
16. **Kyoto** FRB meeting, **Yukawa Institute for Theoretical Physics** (Feb 2021)
15. Compact object group meeting, **TIFR**, Mumbai (Dec 2020)
14. 62nd Annual Meeting of the **APS Division of Plasma Physics** (Nov 2020)
13. Compact objects group meeting, **CCA, Flatiron Institute**, NY (June, Oct, Nov 2020)
12. Plasma astrophysics summer program, **CCA, Flatiron Institute**, NY (July 2019)
11. Theoretical High Energy Astrophysics seminar, **Columbia University** (June 2019)
10. Graduate Seminar, **Columbia University**, NY (Feb, April 2019)
9. Research Seminar, **Columbia University**, NY (Dec 2018)
8. Pizza lunch talk, **Columbia University**, NY (Oct 2018,2022)
7. *NuSTAR* SOC group meeting, **Caltech**, CA (Sept 2017)
6. Group presentation at **NRAO Green Bank Observatory**, WV (Sept 2017)
5. 16th AAS-High Energy Astrophysics Division meeting, **Sun Valley**, ID (March 2022)
4. Visiting Student Program ceremony, **University of Alberta**, Canada (July 2017)
3. Lunch talk at **University of Alberta**, Canada (June 2017)
2. ‘Wide band X-ray spectro-temporal studies’, **TIFR**, Mumbai, India (Jan 2017)
1. **IISER Bhopal** Physics department symposium (Nov 2015)

MENTORING Total: 10 students; 4 papers.

At Stanford University:

- *Shanzay Sajid*; **High-school** student, Lahore Grammar School (Fall 2026-)
- *Ningyue Fan*; **Ph.D.** student, Stanford University (Fall 2025-)
- *Christopher Farias*; **undergraduate** student, San Diego State University (Summer 2025)

At Columbia University:

- *Sanya Gupta*; **undergraduate** research projects (2 papers) → Ph.D. student at Stanford University (2022–2025)
- *Kohki Uno*; **Ph.D.** student, Kyoto University (1 paper) → postdoc at Columbia University (Summer 2024)
- *Shifra Mandel*; **graduate** mentor-mentee program (2021–2022)
- *Arnab Kundu*; **masters** thesis student, IISER Kolkata (2021–2022)
- *Akash Vani*; Pune University → Masters at Heidelberg University → Ph.D. student at MPA, Garching (2019–2020)
- *Shuhan Zhang*; Women in Science at Columbia **undergraduate** mentorship program → Ph.D. student at UC San Diego (2021)
- *Bhairavi Chandrasekhar*; **undergraduate** summer internship (2020)

TEACHING	At Columbia University:	
	– Head TA of the Astronomy department	(2020–2021)
	– Instructor for undergraduate Intro to Astronomy lab courses	(Fall 2019, Spring 2020)
	– Teaching Assistant for 2 undergraduate (non-major) courses	(Fall 2018, Spring 2019)
	– Guest lectured for ASTR UN1403 (in lieu of Prof. David Helfand)	(Feb 2020)
	– Advanced track in Columbia CTL’s Teaching Development Program (TDP)	(2018–2024)
	– Completed CTL’s Innovative Course Design Seminar	(2021)
	Outside Columbia University:	
	– Selected as a Lead instructor at the Yale Young Global Scholars (YYGS) program	
	Yale University , New Haven, CT	(Summer 2020; postponed due to Covid-19)
PROFESSIONAL LEADERSHIP	– Trained to incorporate inquiry-based learning techniques in STEM education through the ISEE—Professional Development Program (PDP) , UC Santa Cruz	(March–Sept 2019)
	– Designed and taught an inquiry activity on statistics, UC Los Angeles	(Sept 2019)
	• Scientific Organizing Committee of symposium on <i>The birth, life and death of black holes</i> , European Astronomical Society annual meeting, Leiden, The Netherlands	(June 2021)
	At Stanford University:	
	• Postbaccalaureate program admissions committee member	(2025)
	• Ph.D. program admissions committee member	(2024–2025)
	• Started the weekly KIPAC Compact Objects Group meetings	(Oct 2024–)
	At Columbia University:	
	• Organized an eight-part lecture series titled <i>‘Science Policy for Scientists’</i>	(Spring 2023)
	• Department representative, Arts and Sciences Graduate Council	(2022–2023)
PROFESSIONAL SERVICE	• Graduate representative, faculty hiring committee, Astronomy department	(2021–2022)
	• Graduate representative, colloquium committee, Astronomy department	(2022–2023)
	• Graduate admissions committee member, Astronomy department	(2021–2022)
	• Co-convenor of <i>Astrofest</i>	(Sept 2020)
	At IISER Bhopal:	
	• Senate representative, Central Advisory Committee, Students Activity Council	(2016–2017)
	• Elected Science Council secretary	(2014–2016)
	• Convenor of <i>Singularity</i> , the Institute annual science festival	(April 2015)
	• Founder of the Institute <i>Astronomy Club</i>	(2014)
	• Founder of the Institute <i>Quiz Club</i>	(2014)
	• Invited to serve as a referee for:	
	–Journal of Plasma Physics,	
	–Proceedings of the National Academy of Sciences of the USA,	
	–Nature,	
	–The Astrophysical Journal,	
	–Monthly Notices of Royal Astronomical Society,	
	–Physical Review D, X,	
	–Astronomy & Astrophysics.	
	• Invited to serve as a panelist for:	
	–NASA’s Chandra, NICER observatories	
	–National Science Foundation,	
	– <i>AstroSat</i> ,	
	–NCRA-uGMRT,	
	–Observer at Hubble Space Telescope time allocation committee.	
	• Science working group member of:	
	–NewAthena	(2025–)
	–Advanced X-ray Imaging Satellite (AXIS)	(2022–)
	–High-Energy X-ray Probe (HEX-P)	(2022–)
	• DEI committee member, Columbia Arts and Sciences Graduate Council	(2022–2023)
	• Judge , Chambliss poster competition, American Astronomical Society meeting	(2022)
SUMMER SCHOOLS	• Webmaster: (1) Columbia Astronomy Department , (2) Theoretical High-Energy Astrophysics group, Columbia , (3) Simons Collaboration on Extreme Electrodynamics of Compact Sources	
	• Member of American Astronomical Society	(2018–present)
	• Member of New York Academy of Sciences	(2018–2023)

- **Multiscale modeling of astrophysical and space plasmas – 2019**
Center for Computer Astrophysics (CCA), Flatiron Institute, New York, NY
- **Multiple approaches to plasma physics from laboratory to astrophysics – 2019**
Ecole de Physique Des Houches, Les Houches, France
- **NRAO Green Bank Telescope observer training workshop – Fall 2017**
Green Bank Observatory, West Virginia
- **Summer School on Gravitational Wave Astronomy – 2016**
International Center for Theoretical Sciences (ICTS), TIFR, Bangalore
- **Summer school on relativity, black holes and neutron stars – 2015**
Indian Institute of Astrophysics (IIAp), Bangalore

COMPUTING/
SOFTWARE
EXPERIENCE

- Languages and tools: Python, git, FORTRAN, IRAF, gnuplot, DS9, L^AT_EX, *f*tools (HEASoft): XSPEC, QDP, XRONOS
- Supercomputers: NASA Pleiades, DoE-NERSC Cori, NSERC WestGrid, Flatiron Institute HPC cluster, Columbia University Habanero, Terremoto, Ginsburg clusters
- Numerical tools: Particle-In-Cell, Magnetohydrodynamical simulations
- Telescope data handling: *AstroSat*, *Swift*, *RXTE*/PCA, *NuSTAR*, *NICER*, *XMM-Newton*, NCRA-GMRT, NRAO-GBT (Astrid, CLEO)

SCIENCE
OUTREACH

1. Public outreach talk during KIPAC community day, Stanford University (April 2025)
2. Research career in astrophysics for undergraduates at IISER Bhopal (Jan 2025)
3. [KIPAC public outreach talk](#), Stanford University (Dec 2024)
4. Co-organized [astronomy outreach activities](#) at Columbia University for the South African academic achievers program (750 high-school students) (Oct 2022)
5. Talk on introduction/orientation to summer research, for undergraduate researchers at Columbia University (June 2022)
6. Popular talk on black hole X-ray binaries at IISER Bhopal Astronomy club (Oct 2021)
7. [Mentored the students of Democracy Prep High School](#), Harlem, NYC, on the foundations of computer coding (Fall 2021)
8. Research career in astrophysics for undergraduates at IISER Bhopal (April 2021)
9. Delivered a lecture on science with space-based X-ray telescopes, for high-school students at *Frontiers of Science*—the annual outreach program at TIFR, Mumbai. (Nov 2017)
10. Organized a tour of GMRT for TIFR–Visiting Summer Research students (June 2016)

News coverage and appearances:

- *Brightest fast radio burst ever detected could help solve an enduring cosmic mystery* [[CNN](#); 26-Aug-2025]
- *Super sharp images with EVN reveal another Fast Radio Burst Coming from a Hypernebula.* [[JIVE](#); 29-Nov-2023][[ASTRON](#); 01-Dec-2023][[Universe Today](#); 05-Dec-2023]
- *Astronomers caught a potent radio burst blasting at us from a dwarf galaxy 3 billion light-years away* [[Popular Science](#); 09-June-2022]
- *AstroSat Catches Nuclear Reactions Spreading Across a Neutron Star* [[The Wire](#); 07-October-2021]
- *Long burst short: an unexpected supernova* [[Research Matters](#); 26-July-2021]
- *Fast Radio Bursts from globular clusters and what could power them* [[National Geographic](#); 27-May-2021]
- *Digging the grave: In search of the remains of merging black holes and neutron stars* [[Research Matters](#); 14-Sept-2021]
- *FRBs from Galactic magnetar* [[The Wire](#); 03-June-2020] [[AAS Nova](#); 09-Sept-2020]
- *Black hole bends light on itself* [[Gizmodo](#); 10-April-2020] [[Caltech News](#); 08-April-2020] [[SYFY](#); 09-July-2020]